AWS SNS, SQS and SES Beginner Tasks

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Amazon SNS

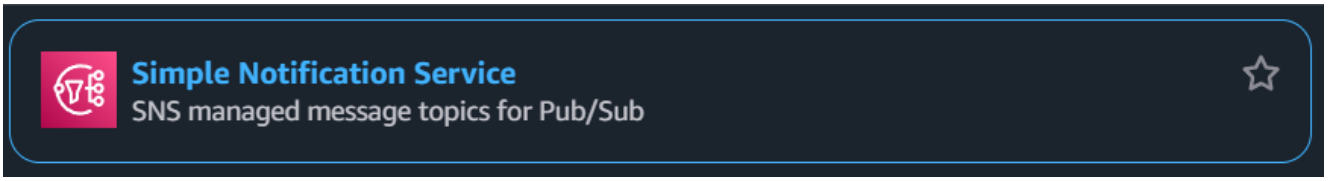
Basic Terms

- ARN (Amazon Resource Name) is a unique ID that AWS gives to every resource, like an ID card.
- Amazon SNS (Simple Notification Service) sends messages to subscribers such as email addresses or other AWS services.

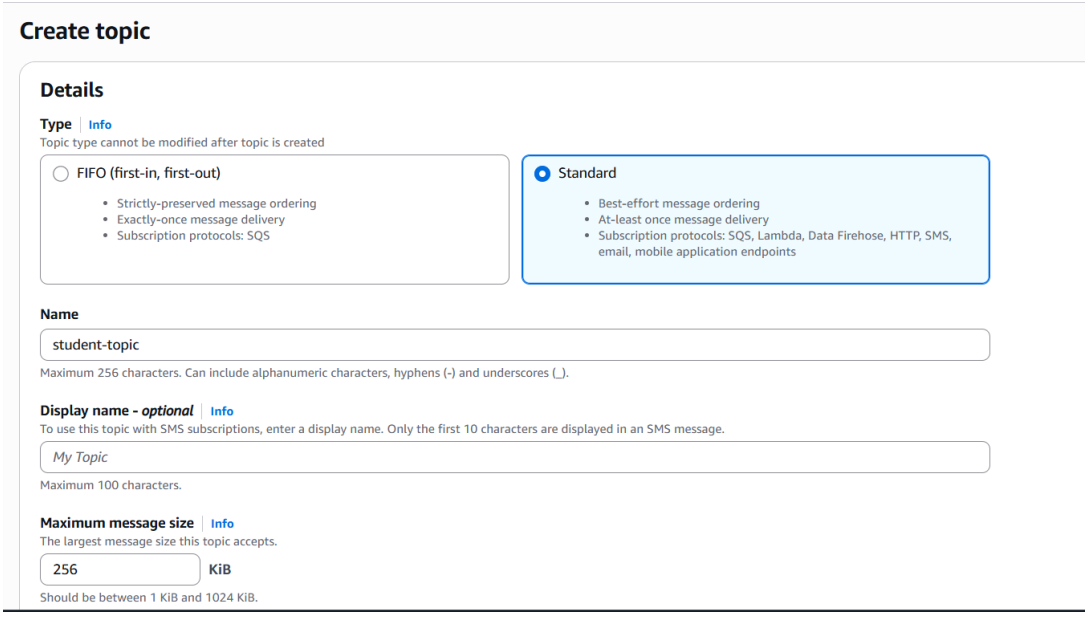
Task 1: Create a Standard SNS topic named student-topic and view its ARN

- Opened Amazon SNS, clicked Create topic, selected type Standard and entered the name student-topic.
- Topic ARN: arn:aws:sns:ap-southeast-2:942440205010:student-topic

Screenshot 1: Opening Amazon SNS service



Screenshot 2: Create topic page (Standard, student-topic)



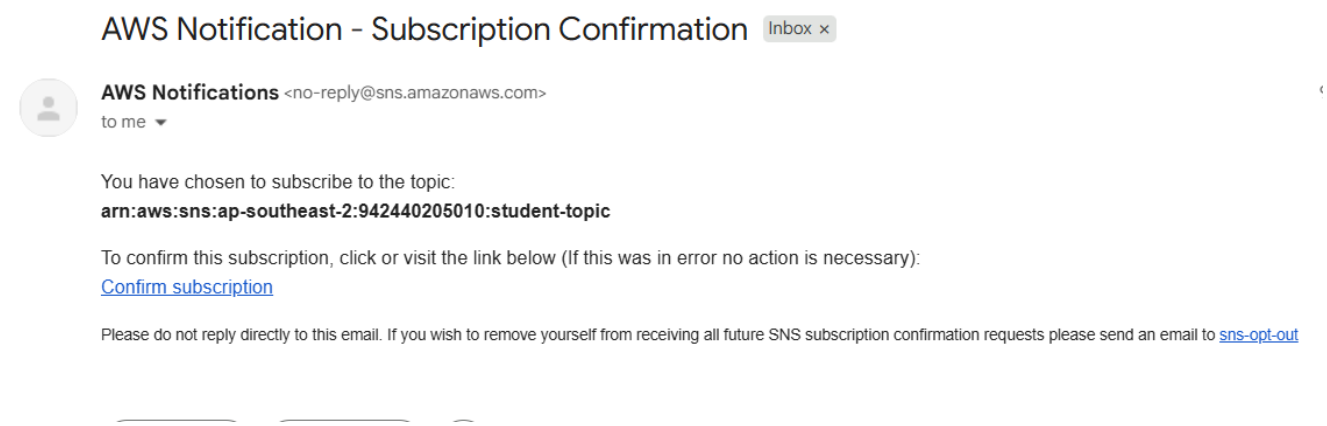
Task 2: Subscribe your email and confirm the subscription

- Created a subscription with protocol Email and endpoint abduljaneeb@gmail.com. The status was Pending confirmation.
- Clicked Confirm subscription in the AWS email, and the status changed to Confirmed.

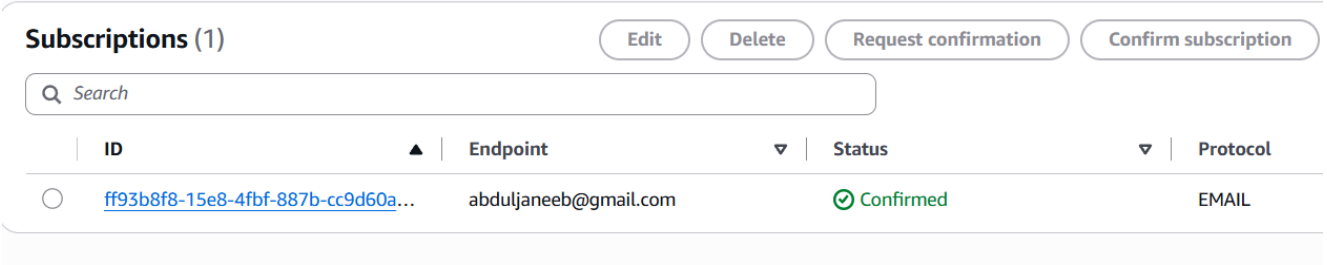
Screenshot 3: Subscription created (Pending confirmation)



Screenshot 4: Confirmation email from AWS



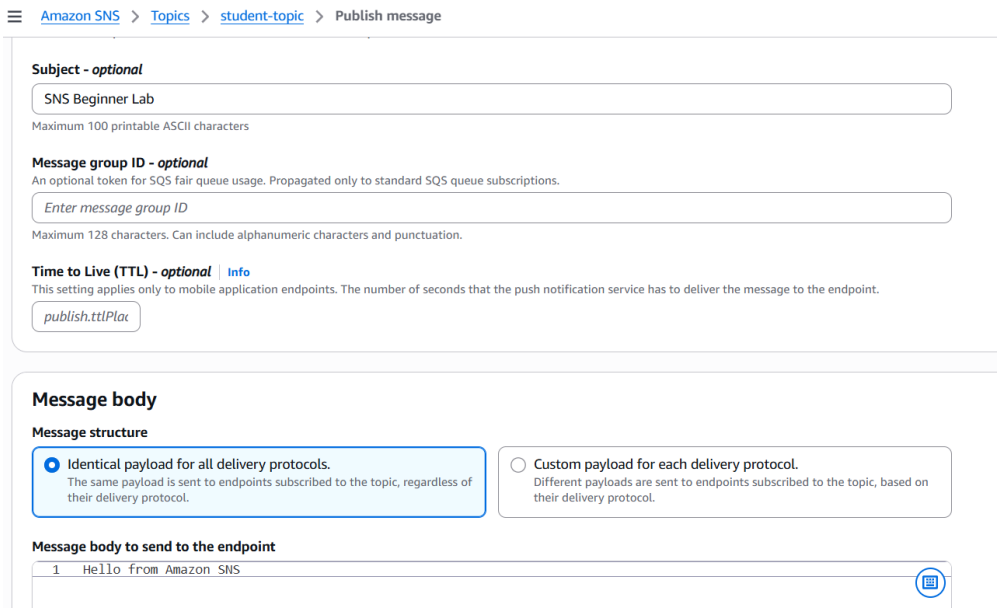
Screenshot 5: Subscription status Confirmed



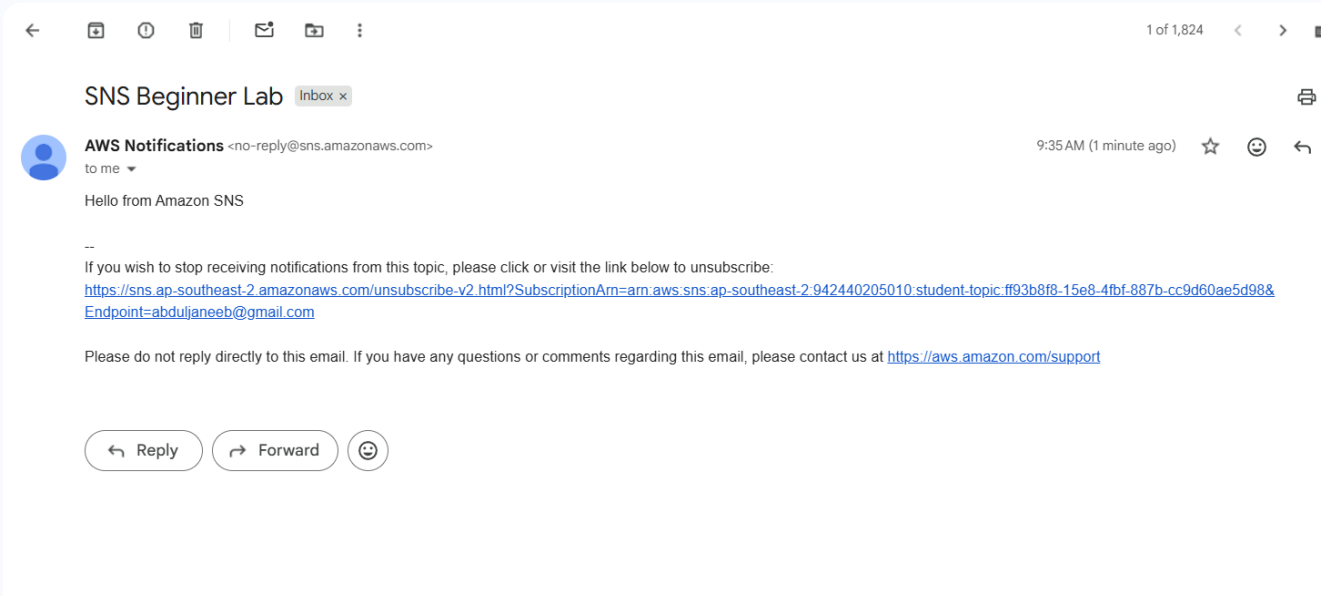
Task 3: Publish a message 'Hello from Amazon SNS' and verify email delivery

- Opened student-topic, clicked Publish message, set Subject as SNS Beginner Lab and message body as Hello from Amazon SNS.
- The email was received in the inbox from AWS Notifications, so delivery is verified.

Screenshot 6: Publish message page



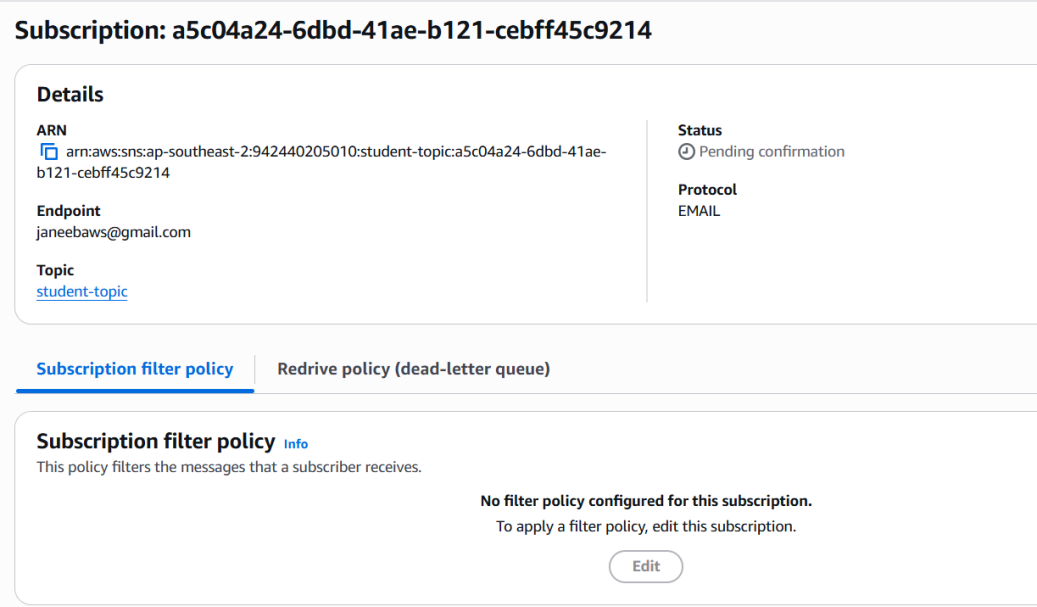
Screenshot 7: Email received in Gmail



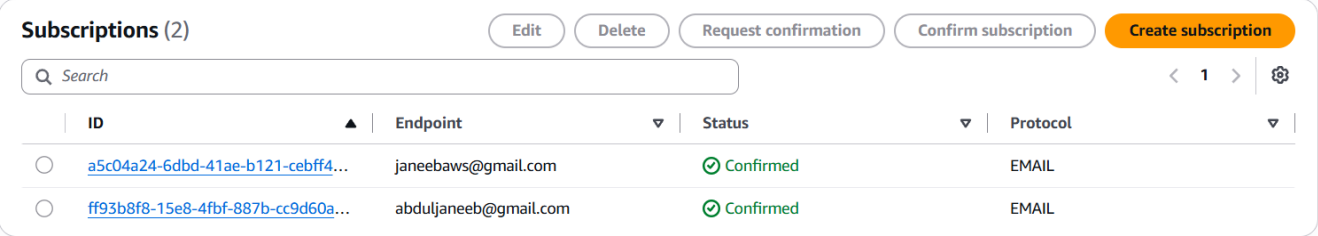
Task 4: Add another email subscriber and publish another message

- Added a second subscriber, janeebaws@gmail.com, and confirmed it. The topic now has 2 confirmed subscriptions.
- Published a new message (subject: SNS Task 4 Test) and the second email address received it.

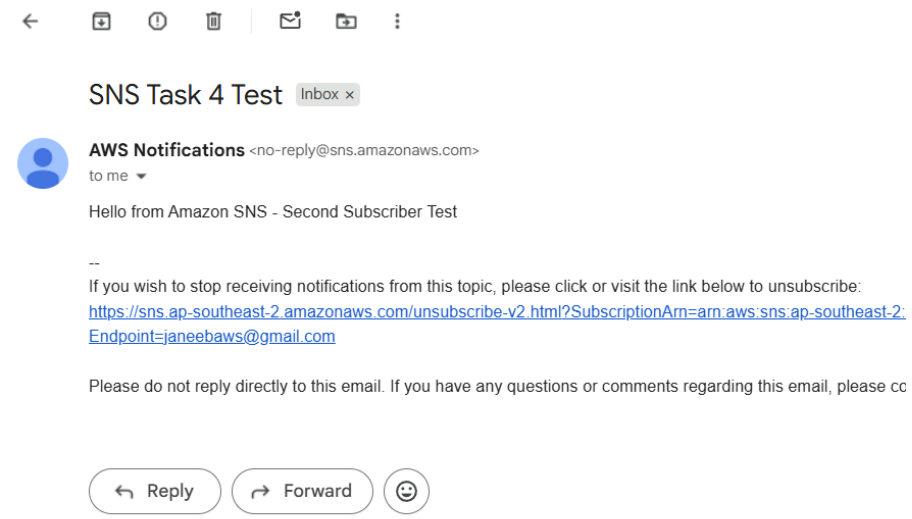
Screenshot 8: Second subscriber added (Pending confirmation)



Screenshot 9: Both subscriptions Confirmed



Screenshot 10: Second subscriber received the email



Task 5: Publish a message using the AWS CLI

- Used AWS CloudShell and ran the command below. It returned a MessageId, which means the message was published.
- The email 'Hello from AWS CLI' was received by the subscriber.

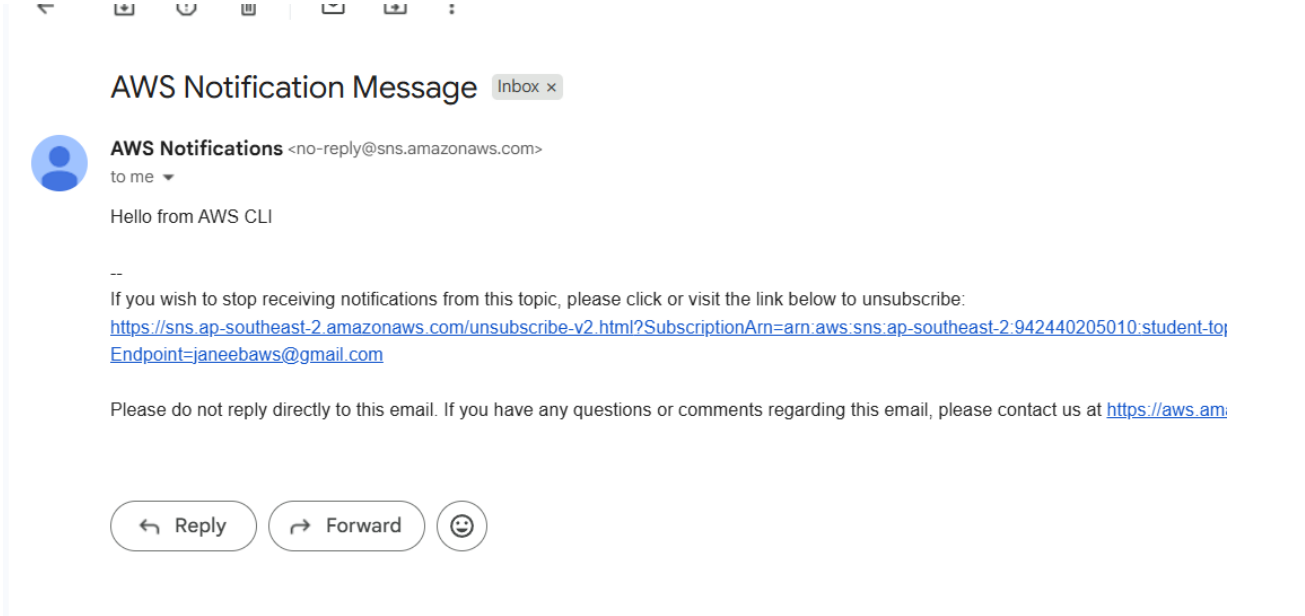
```
aws sns publish \
```

```
--topic-arn arn:aws:sns:ap-southeast-2:942440205010:student-topic \
--message "Hello from AWS CLI"
```

Screenshot 11: AWS CLI publish command in CloudShell



Screenshot 12: Email received from CLI message



Amazon SQS

Basic Terms

- Amazon SQS (Simple Queue Service) is an AWS service that stores messages in a queue until an application retrieves and processes them.
- A queue keeps the sender and receiver separate, so messages are not lost even if the receiver is busy.

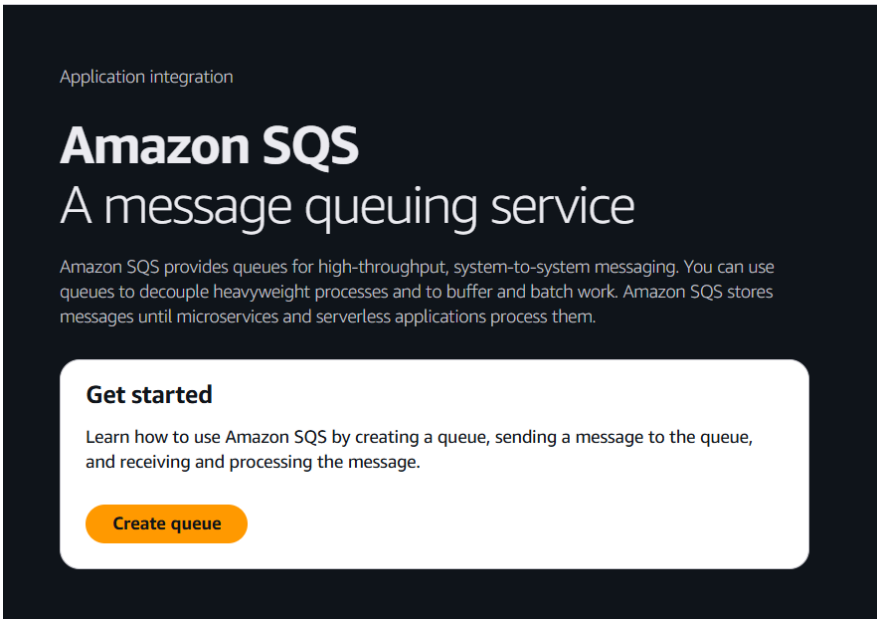
Task 1: Create a Standard queue named student-queue

- Opened Amazon SQS, clicked Create queue, selected type Standard and entered the name student-queue.
- Queue ARN: arn:aws:sqs:ap-southeast-2:942440205010:student-queue

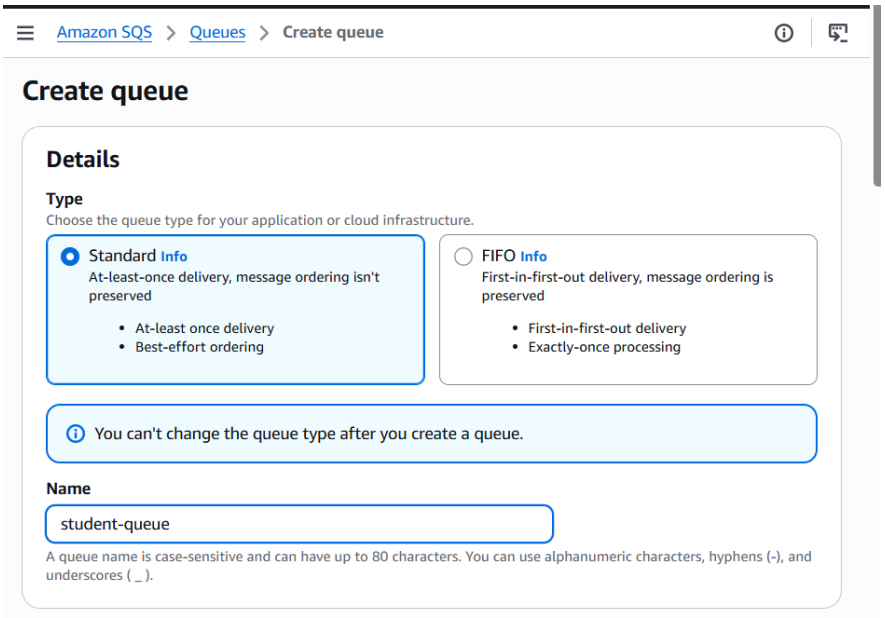
Screenshot 1: Opening Amazon SQS service



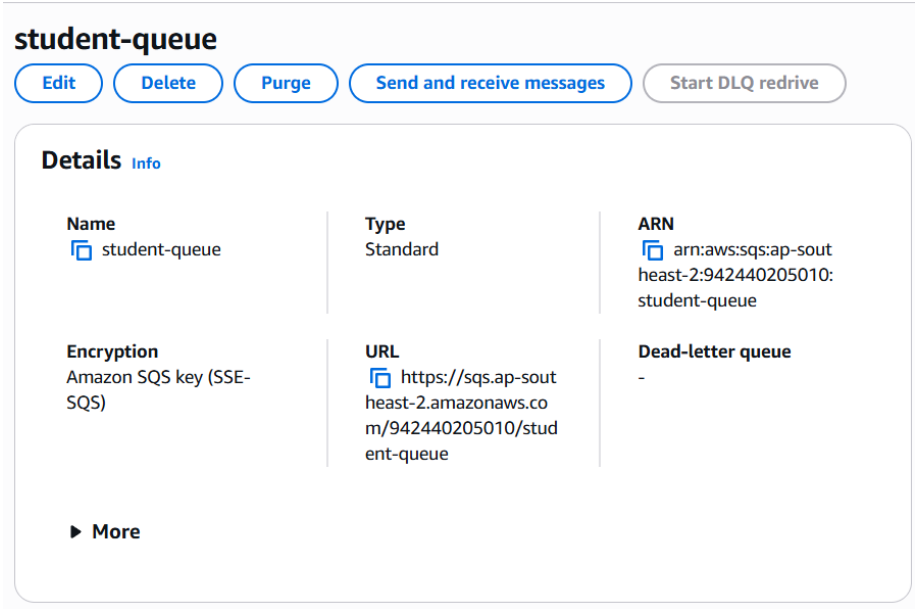
Screenshot 2: Amazon SQS home page (Create queue)



Screenshot 3: Create queue page (Standard, student-queue)



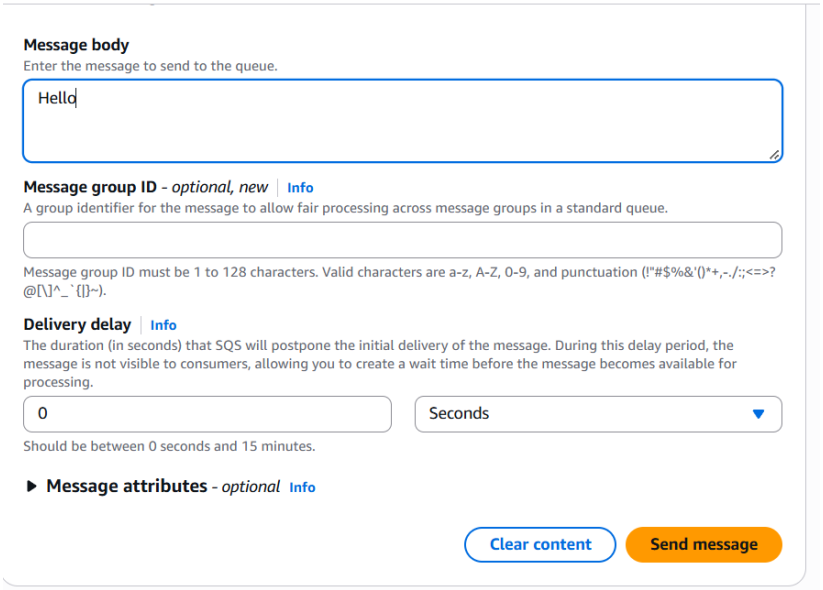
Screenshot 4: student-queue created (details and ARN)



Task 2: Send three messages: Hello, AWS, SQS

- Opened student-queue, clicked Send and receive messages, typed the message body and clicked Send message.
- Repeated this for the three messages: Hello, AWS and SQS.

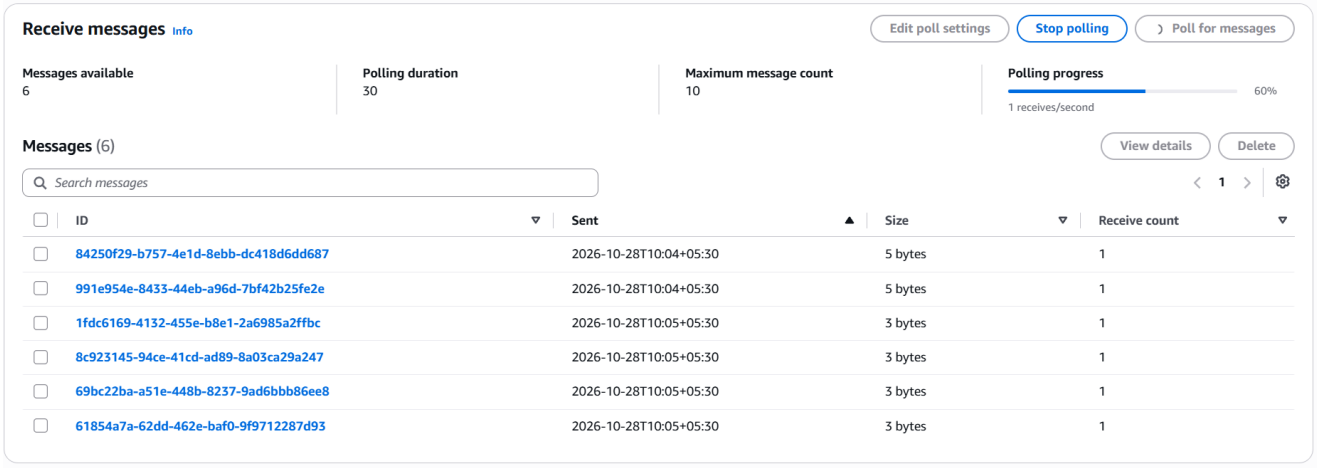
Screenshot 5: Sending a message to the queue



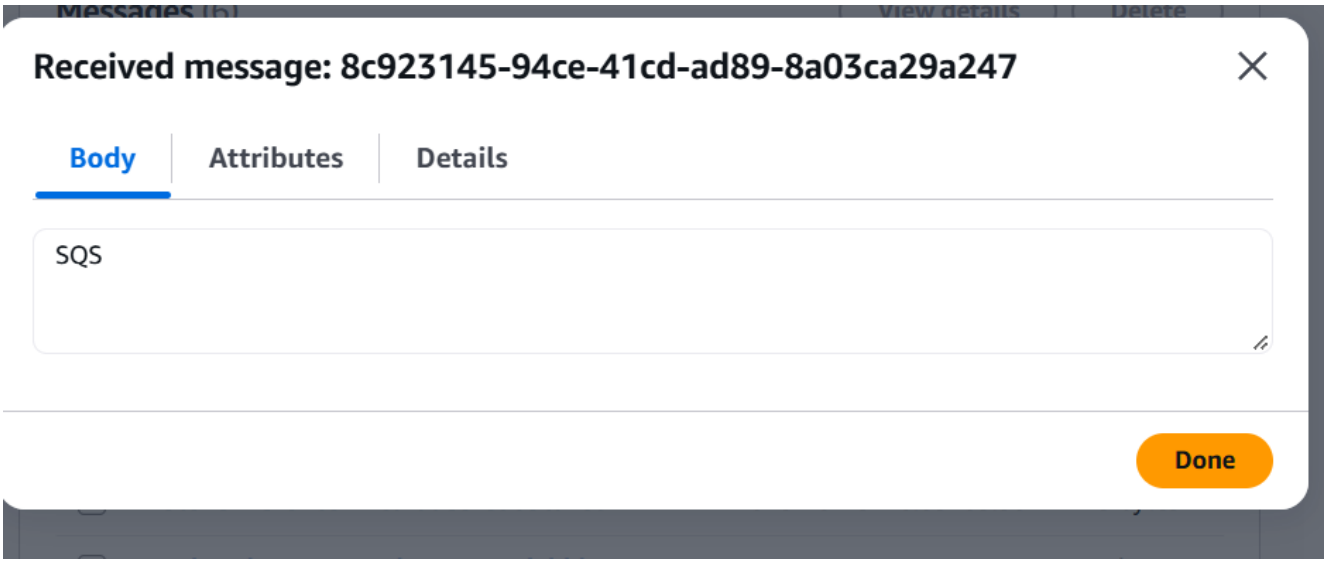
Task 3: Receive the messages from the console

- Clicked Poll for messages, and the sent messages appeared in the Messages list with their IDs.
- Opened a message using View details to read its body (for example, SQS).

Screenshot 6: Messages received after polling



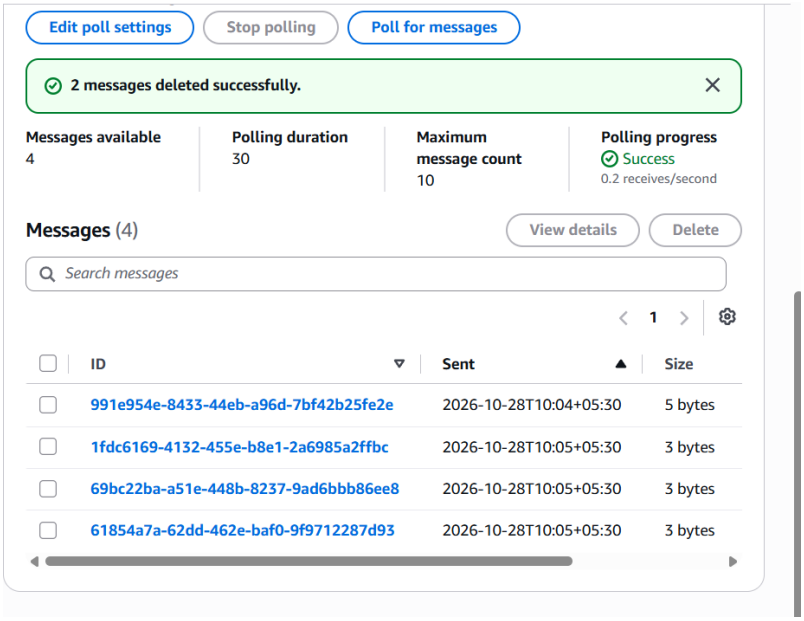
Screenshot 7: Message body shown (SQS)



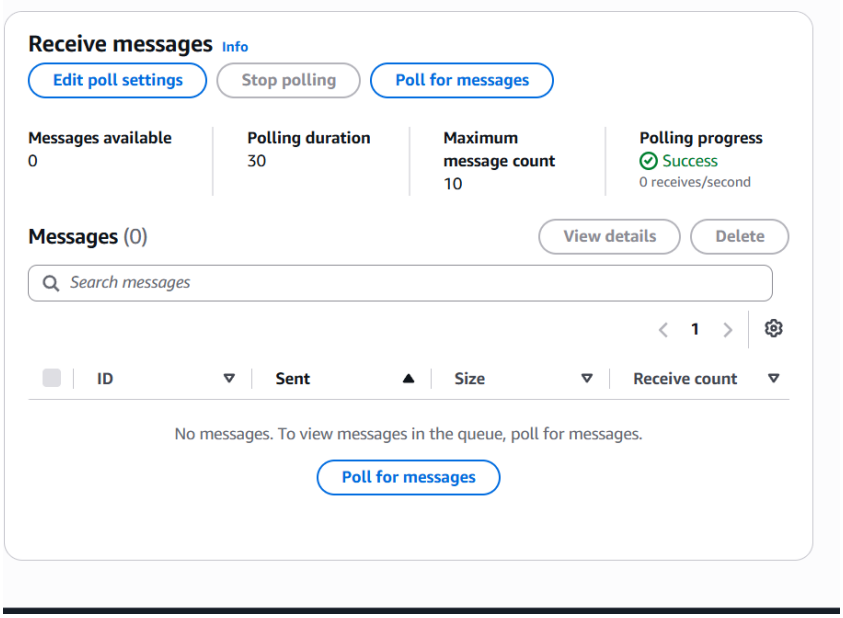
Task 4: Delete the messages and verify the queue is empty

- Selected the messages and clicked Delete. AWS showed a message deleted successfully.
- After deleting all messages, Messages available shows 0, so the queue is empty.

Screenshot 8: Messages deleted successfully



Screenshot 9: Queue is empty (0 messages)



Task 5: Send a message using the AWS CLI and receive it from the console

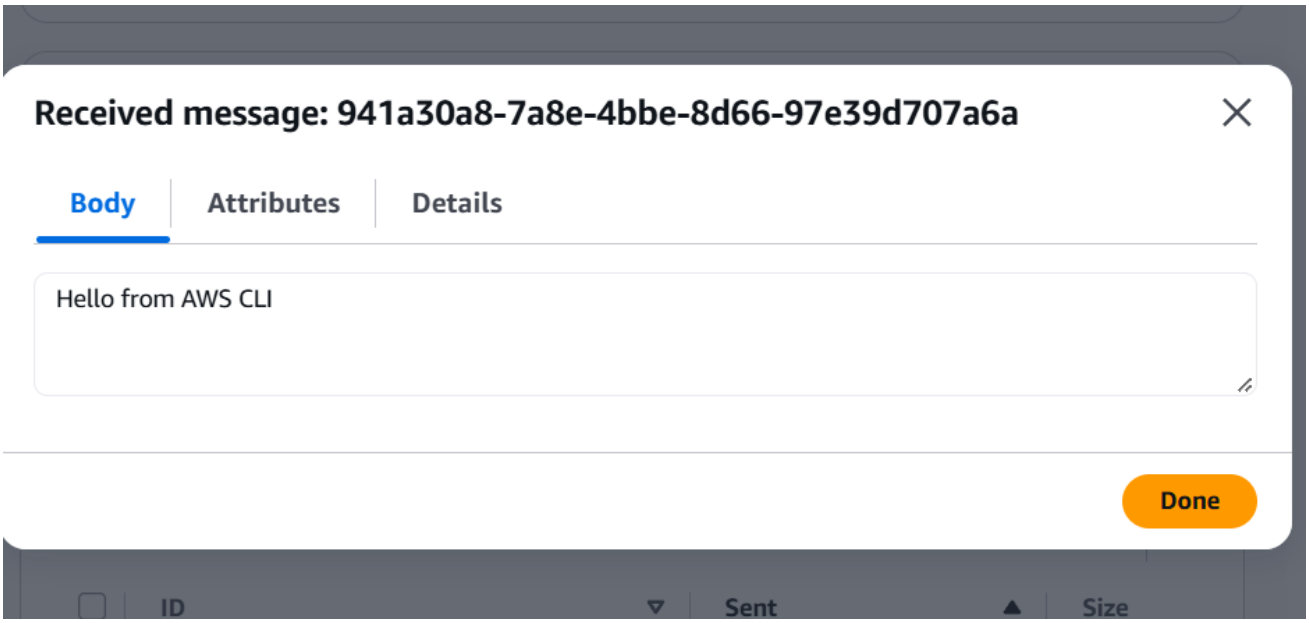
- Used AWS CloudShell and ran the command below. It returned a MessageId, which means the message was sent.
- Polled for messages in the console and the message 'Hello from AWS CLI' was received.

```
aws sqs send-message \
  --queue-url "https://sqs.ap-southeast-2.amazonaws.com/942440205010/student-queue" \
  --message-body "Hello from AWS CLI"
```

Screenshot 10: AWS CLI send-message command in CloudShell



Screenshot 11: Message received in the console



Amazon SES

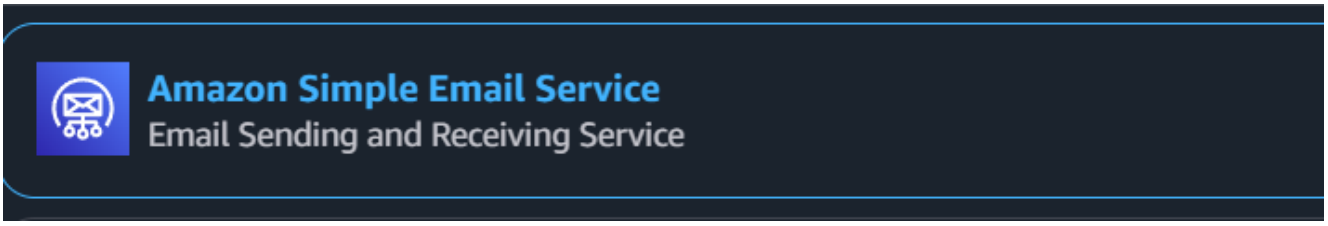
Basic Terms

- Amazon SES (Simple Email Service) is an AWS service used to send and receive emails.
- In the SES sandbox, emails can only be sent between verified email addresses.

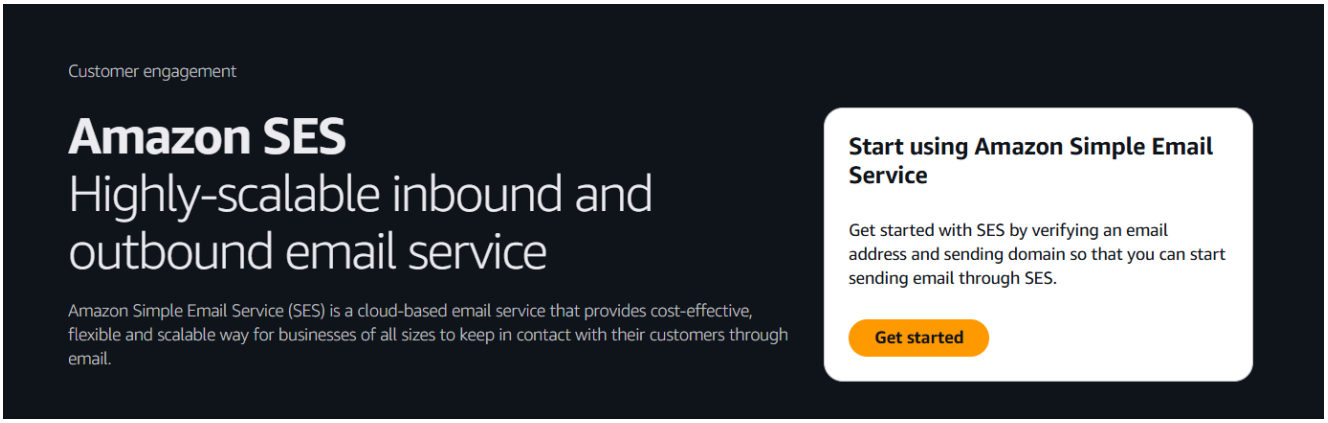
Task 1: Verify an email address

- Opened Amazon SES, created an identity of type Email address with abduljaneeb@gmail.com, and clicked the verification link sent to the inbox.
- Both abduljaneeb@gmail.com and janeebaws@gmail.com now show the status Verified.

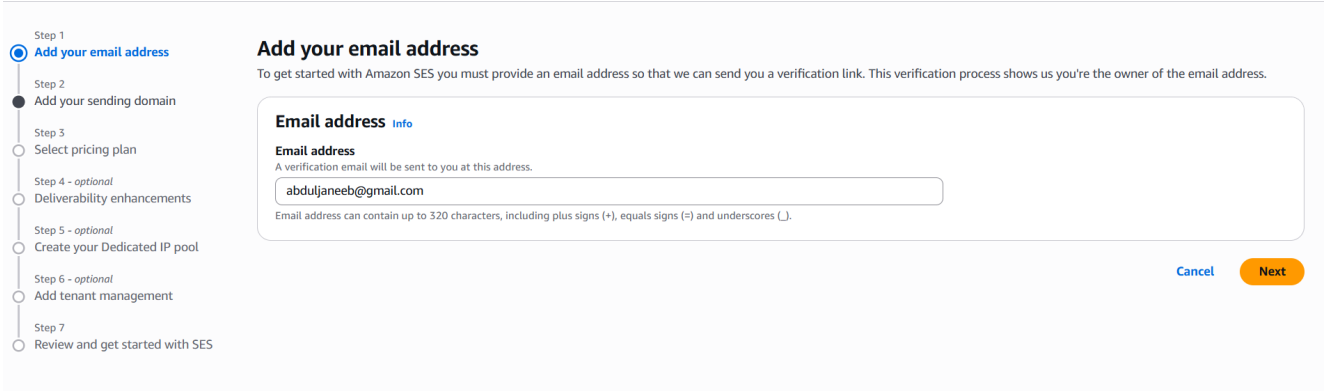
Screenshot 1: Opening Amazon SES service



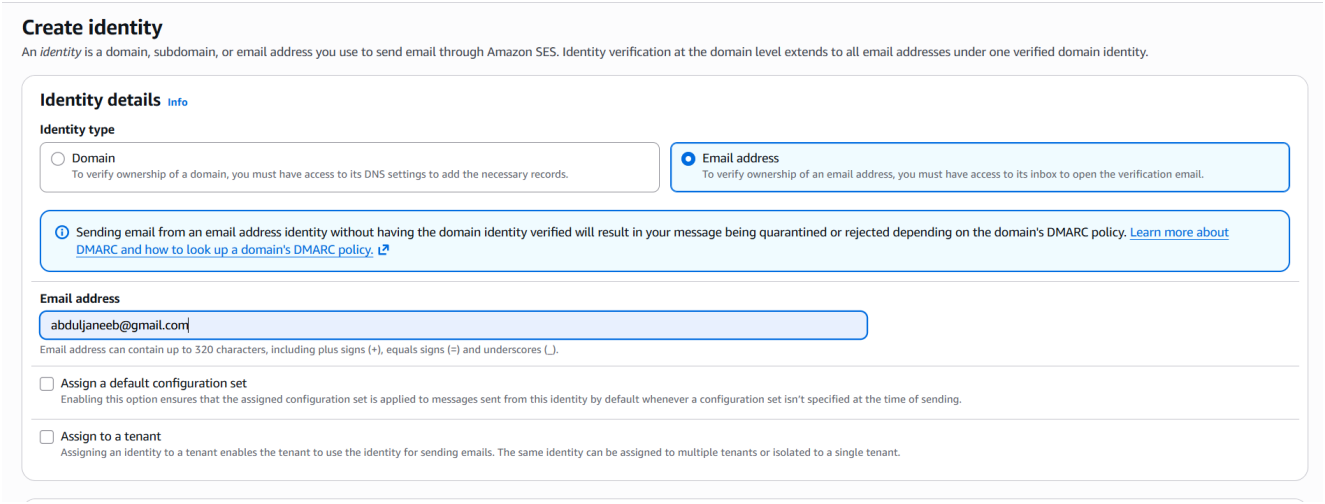
Screenshot 2: Amazon SES home page



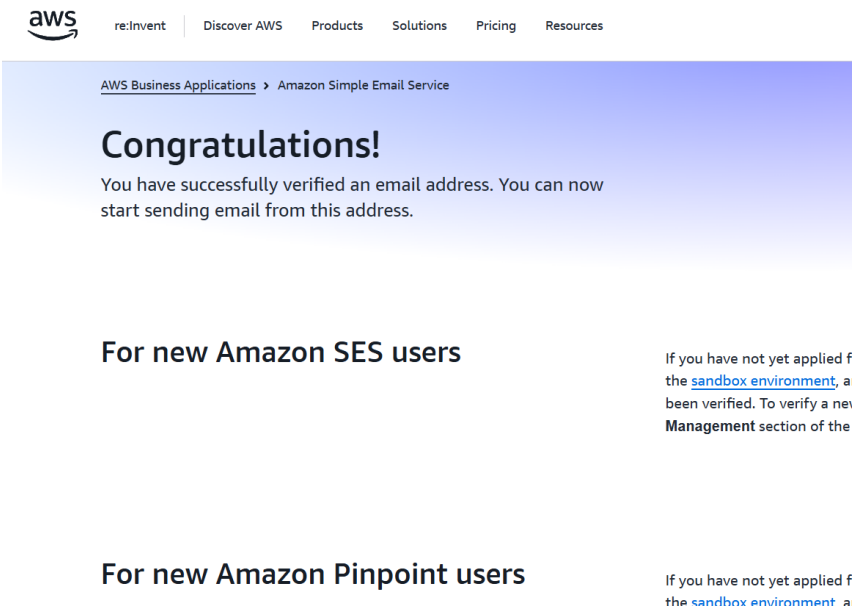
Screenshot 3: Adding the email address



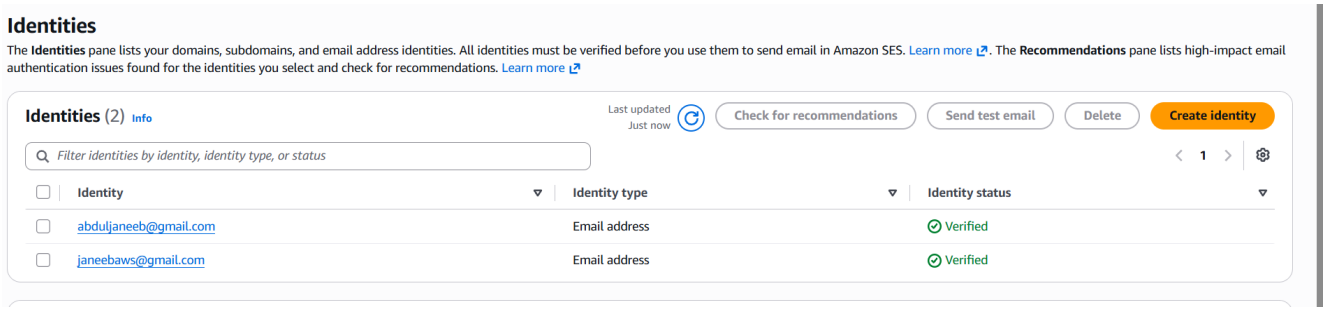
Screenshot 4: Create identity (Email address)



Screenshot 5: Email verified successfully



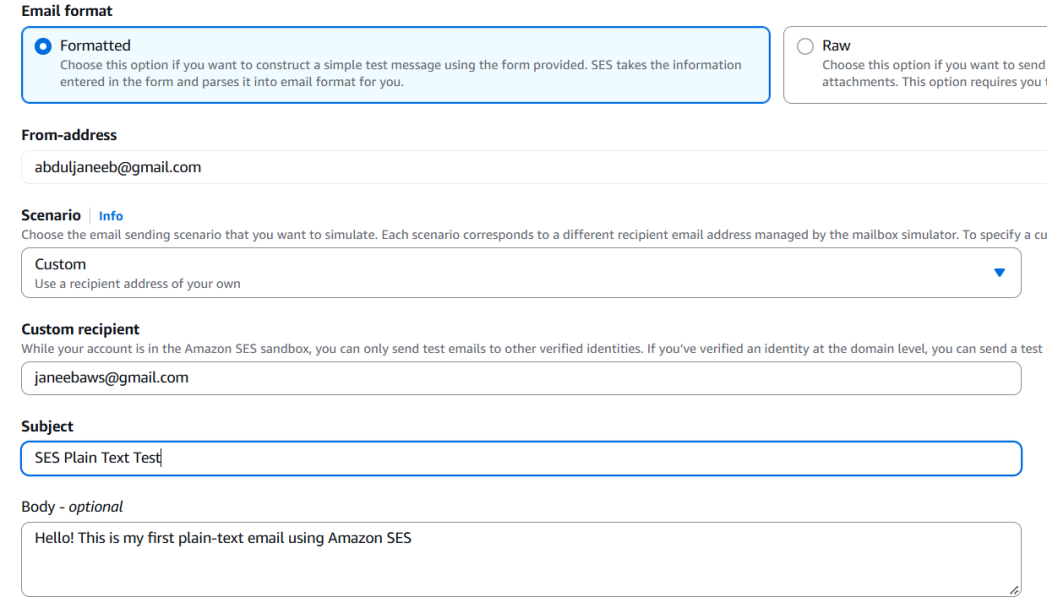
Screenshot 6: Identities list showing Verified



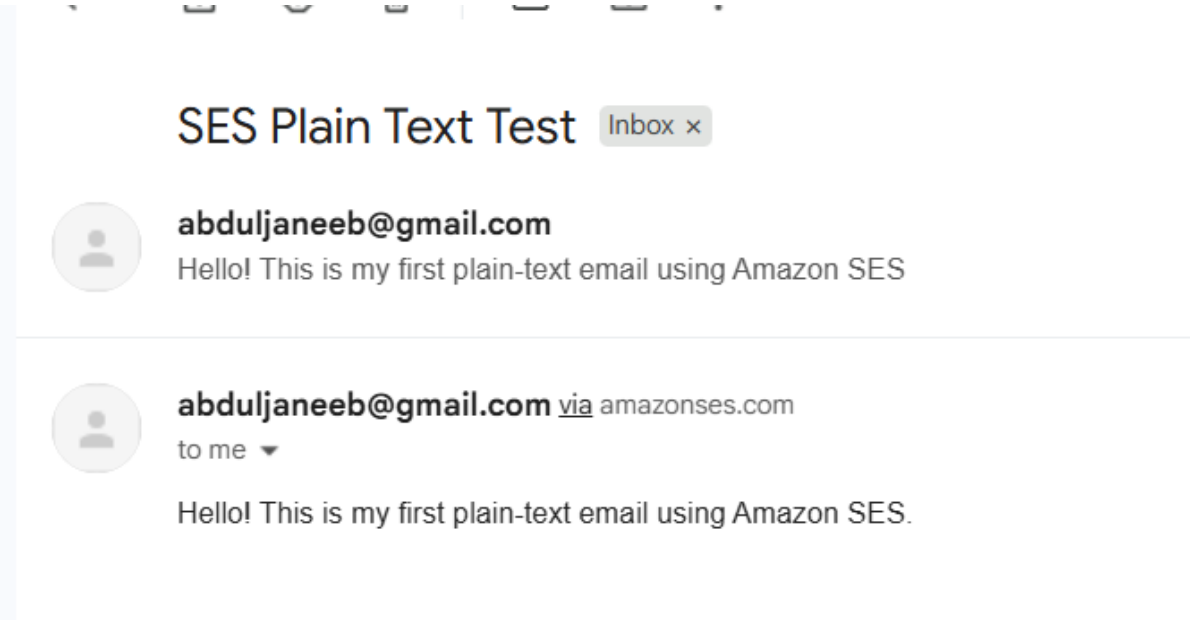
Task 2: Send a plain text email between verified addresses

- Used Send test email with the Formatted option, from abduljaneeb@gmail.com to janeebaws@gmail.com, subject SES Plain Text Test.
- The plain text email was received in the inbox, so it was delivered successfully.

Screenshot 7: Send test email (Formatted)



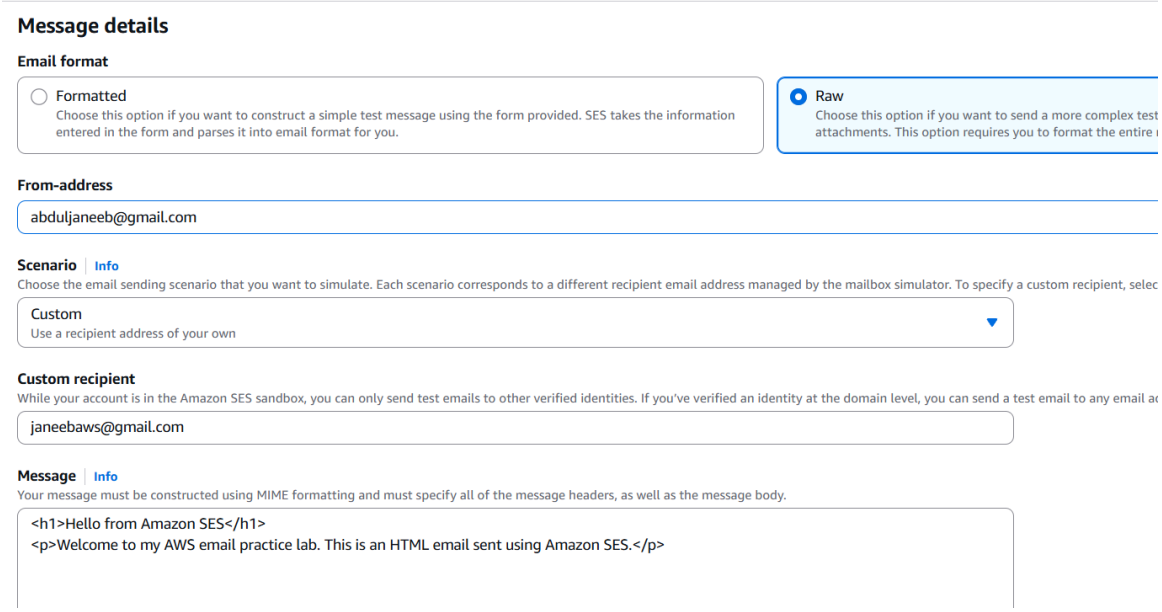
Screenshot 8: Plain text email received



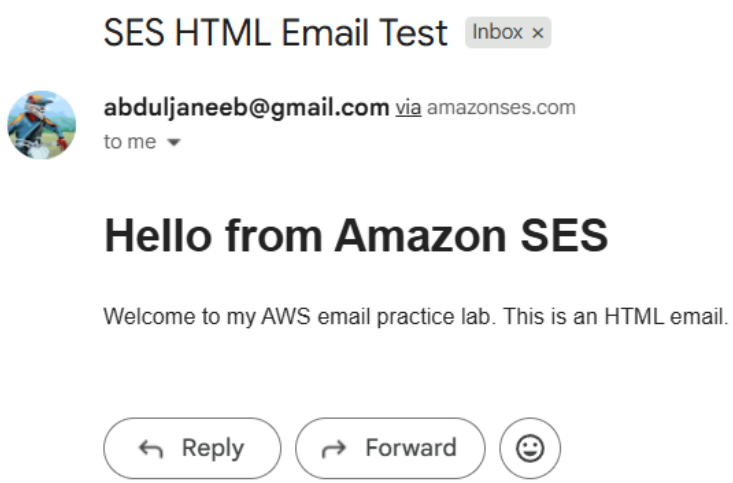
Task 3: Send an HTML email with a heading and paragraph

- Used Send test email with the Raw option and wrote HTML with an h1 heading and a p paragraph.
- The email was received with a large heading and a paragraph below it, so the HTML worked.

Screenshot 9: Send test email (Raw HTML)



Screenshot 10: HTML email received



Task 4: Send an email using the AWS CLI

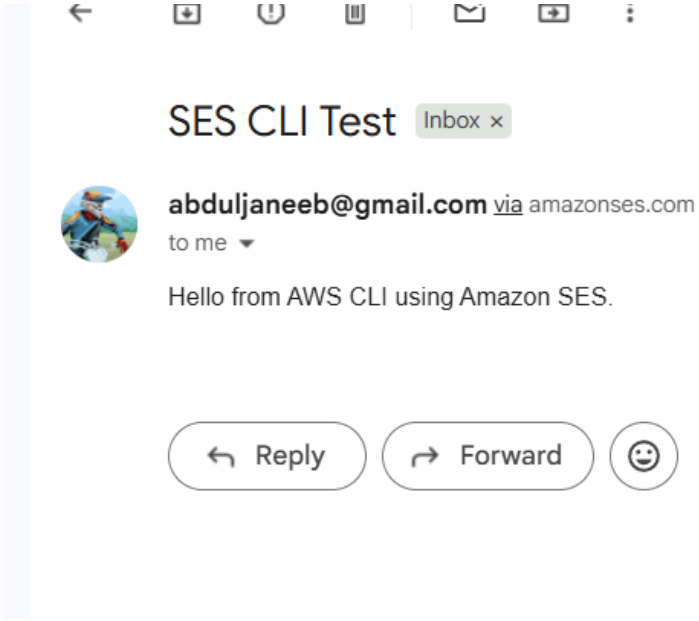
- Used AWS CloudShell and ran the command below. It returned a MessageId, which means the email was sent.
- The email 'SES CLI Test' was received in the inbox.

```
aws ses send-email \
  --from "abduljaneeb@gmail.com" \
  --destination "ToAddresses=janeebaws@gmail.com" \
  --message 'Subject={Data="SES CLI Test"},Body={Text={Data="Hello from AWS CLI using Amazon SES."}}'
```

Screenshot 11: AWS CLI send-email command in CloudShell



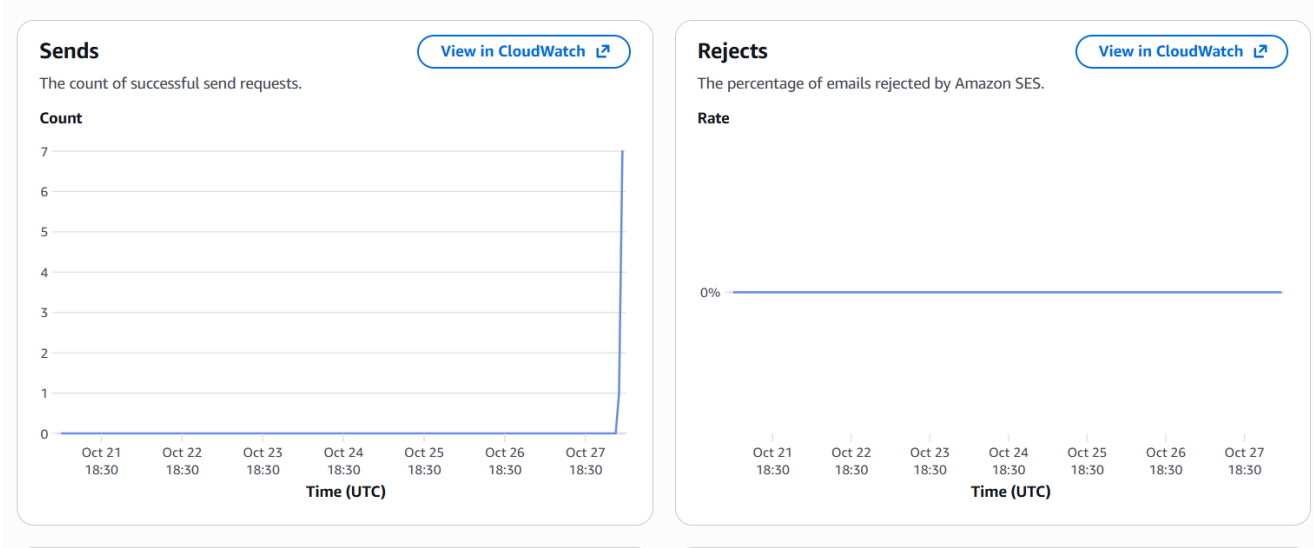
Screenshot 12: CLI email received



Task 5: View sending statistics in the SES dashboard

- Opened the SES dashboard, where the Sends graph shows the count of successful send requests (7).
- Rejects, Bounces and Complaints are all 0%, which means no emails failed or were marked as spam.

Screenshot 13: Sends and Rejects



Screenshot 14: Bounces and Complaints

